

JEE Mains 2019 Chapter wise Question Bank

Circular Motion - Questions

Q1

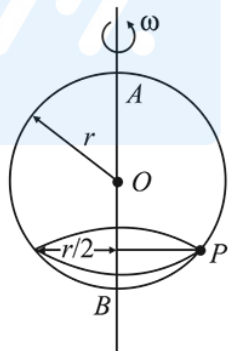
A particle is moving along a circular path with a constant speed of 10 ms^{-1} . What is the magnitude of the change in velocity of the particle, when it moves through an angle of 60° around the centre of the circle?

- (1) $10\sqrt{3} \text{ m/s}$ (2) zero
(3) $10\sqrt{2} \text{ m/s}$ (4) 10 m/s

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Q2

A smooth wire of length $2\pi r$ is bent into a circle and kept in a vertical plane. A bead can slide smoothly on the wire. When the circle is rotating with angular speed ω about the vertical diameter AB, as shown in figure, the bead is at rest with respect to the circular ring at position P as shown. Then the value of ω^2 is equal to :



- (1) $\frac{\sqrt{3}g}{2r}$ (2) $2g / (r\sqrt{3})$
(3) $(g\sqrt{3}) / r$ (4) $2g/r$

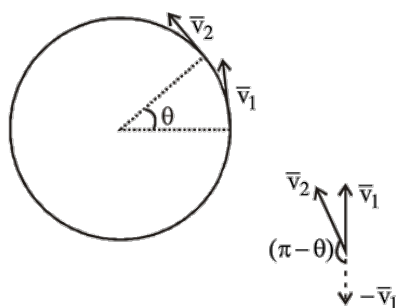
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Circular Motion – Answers

Q1

(4)



Change in velocity,

$$|\Delta \vec{v}| = \sqrt{v_1^2 + v_2^2 + 2v_1 v_2 \cos(\pi - \theta)}$$

$$= 2v \sin \frac{\theta}{2} \quad (\because |\vec{v}_1| = |\vec{v}_2| = v)$$

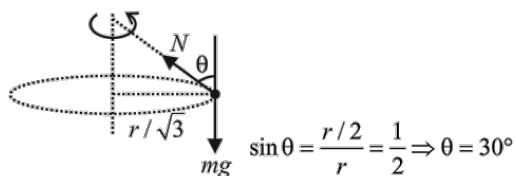
$$= (2 \times 10) \times \sin(30^\circ) = 2 \times 10 \times \frac{1}{2}$$

$$= 10 \text{ m/s}$$

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Q2

$$(2) N \sin \theta = m\omega^2 (r/2) \quad \dots(i)$$



$$\text{and } N \cos \theta = mg \quad \dots(ii)$$

$$\text{or } \tan \theta = \frac{\omega^2 r}{2g}$$

$$\text{or } \tan 30^\circ = \frac{\omega^2 r}{2g}$$

$$\text{or } \frac{1}{\sqrt{3}} = \frac{\omega^2 r}{2g}$$

$$\therefore \omega^2 = \frac{2g}{r\sqrt{3}}$$

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